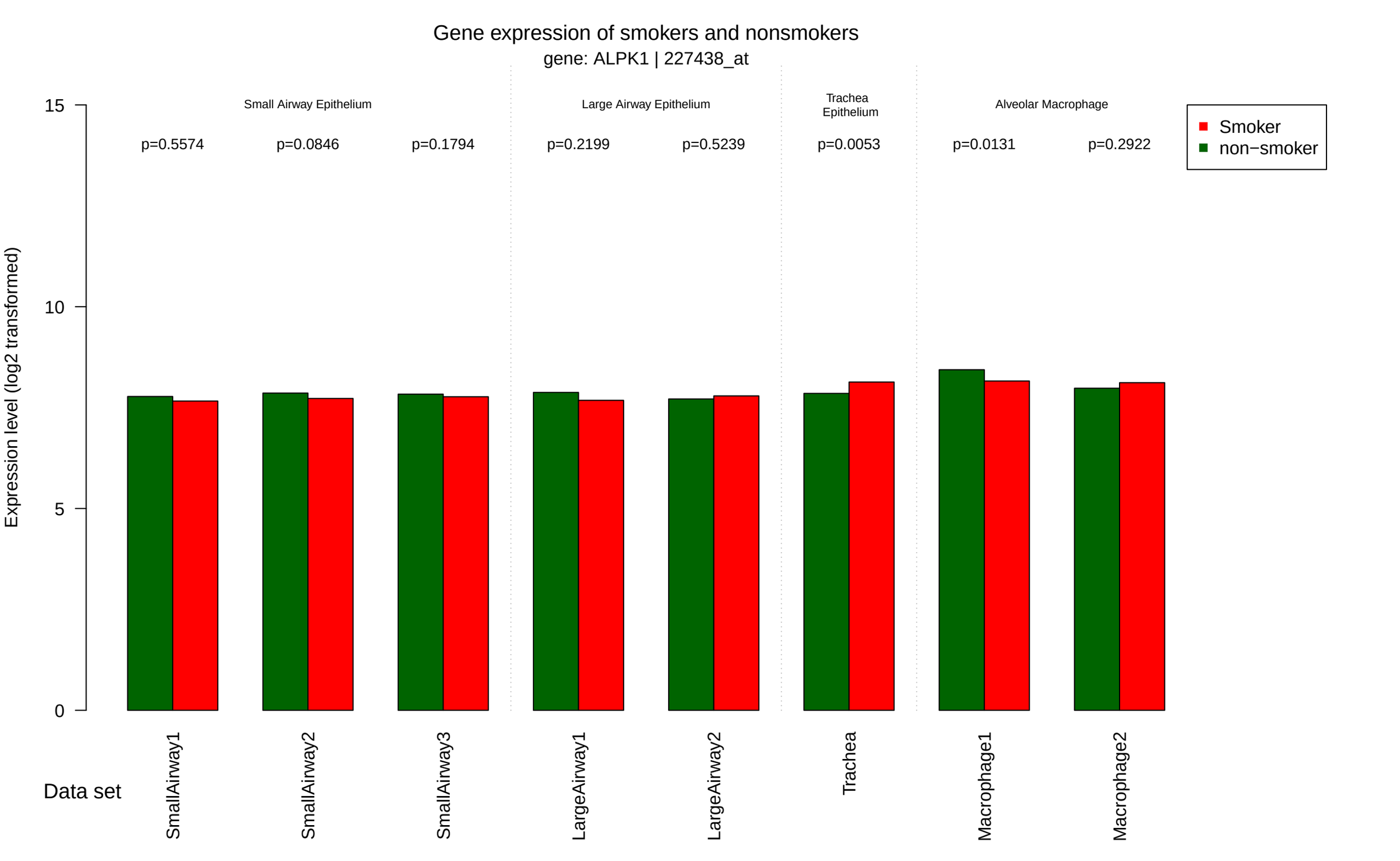
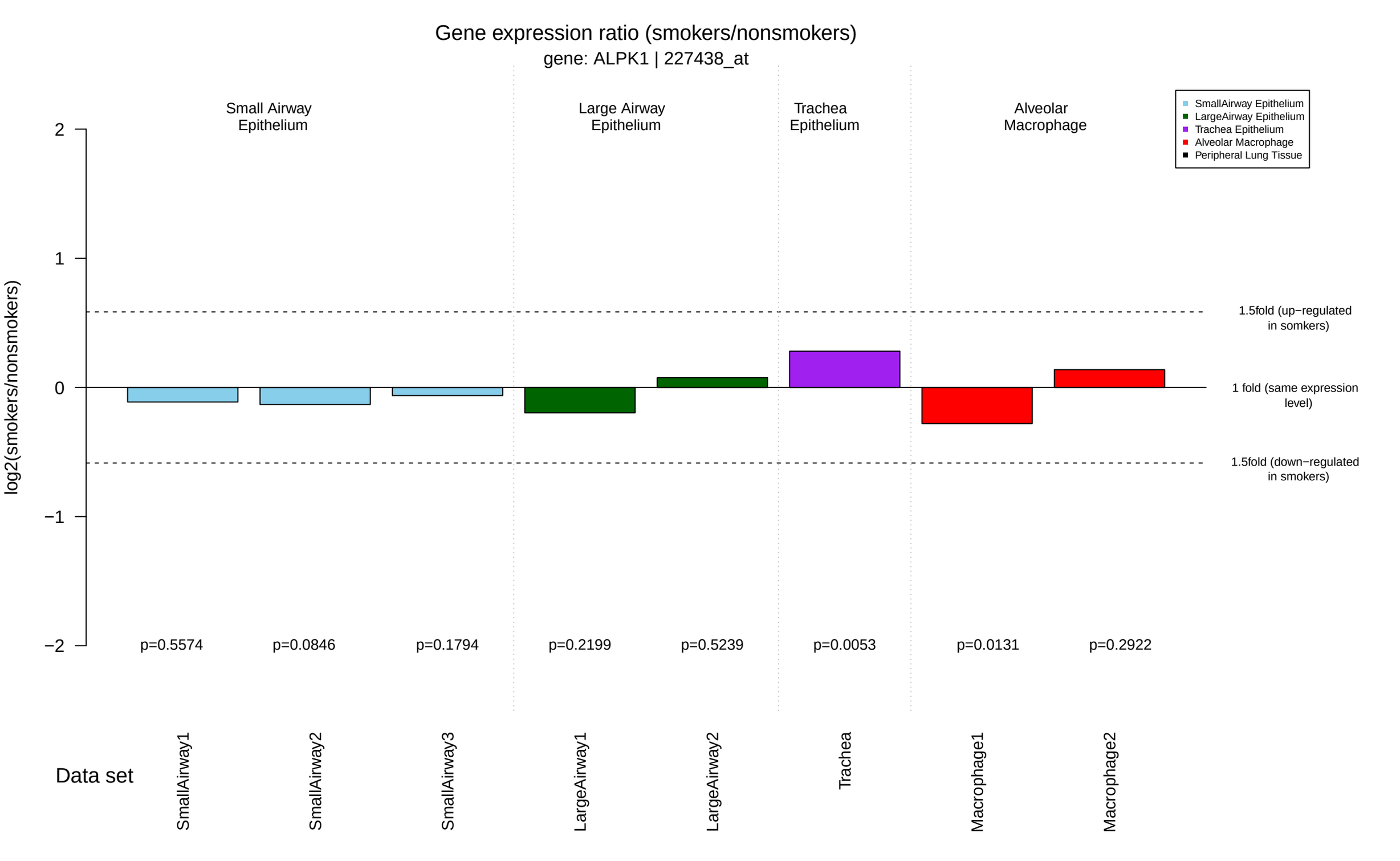
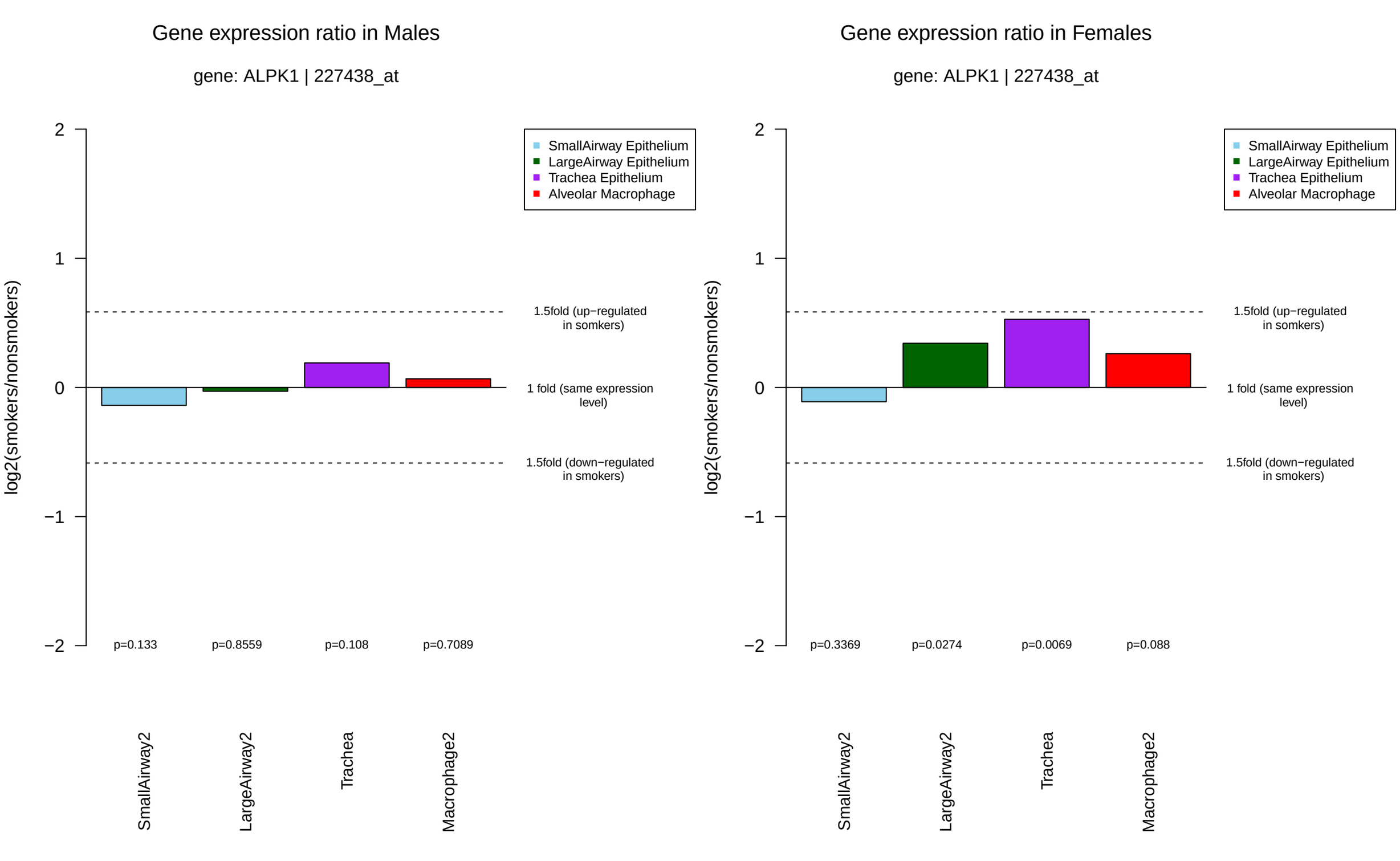
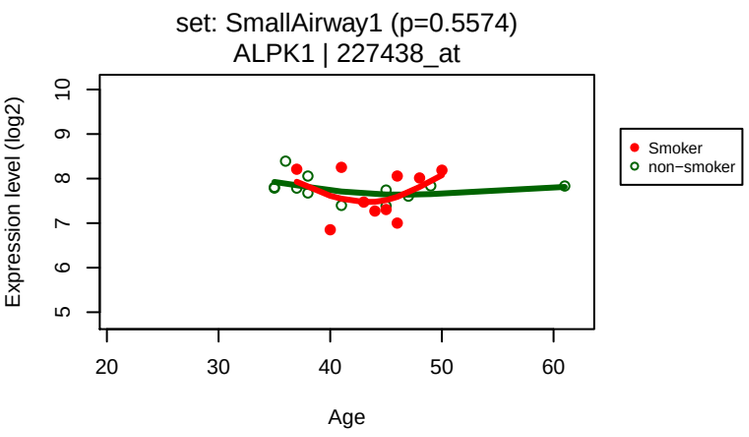
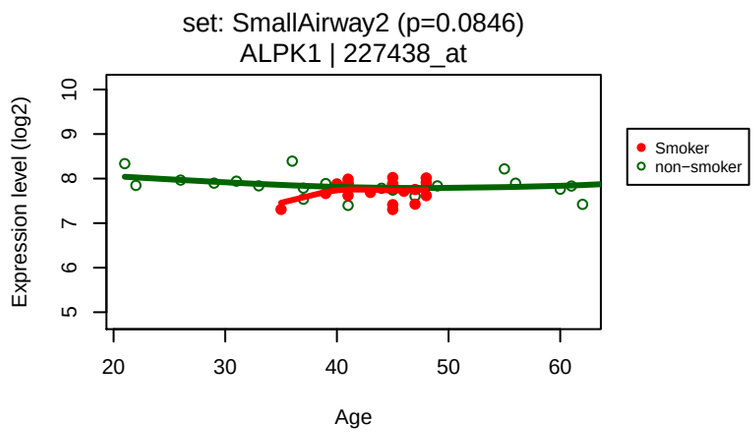
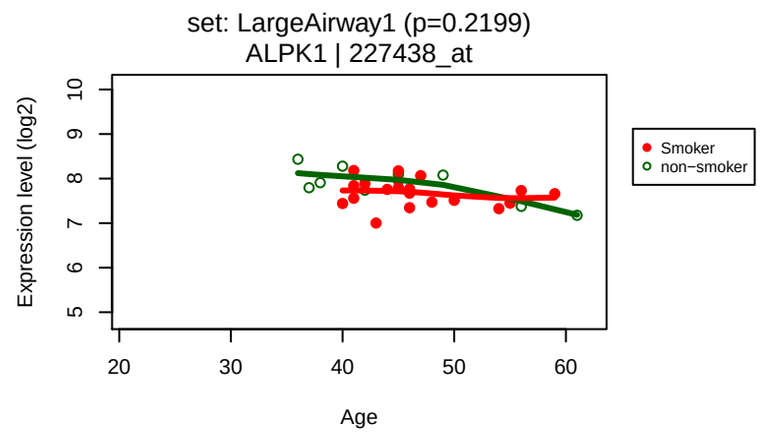

Gene expression of smokers and nonsmokers
gene: ALPK1 | 227438_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5574
p=0.0846
p=0.1794
p=0.2199
p=0.5239
p=0.0053
p=0.0131
p=0.2922
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

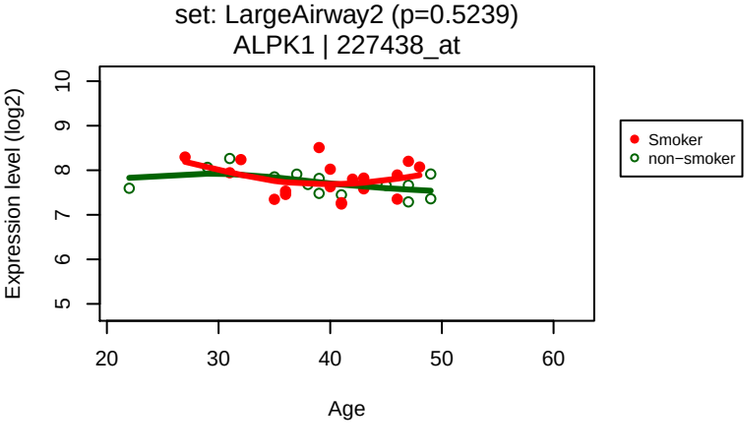

Gene expression ratio (smokers/nonsmokers)
gene: ALPK1 | 227438_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5574
p=0.0846
p=0.1794
p=0.2199
p=0.5239
p=0.0053
p=0.0131
p=0.2922
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

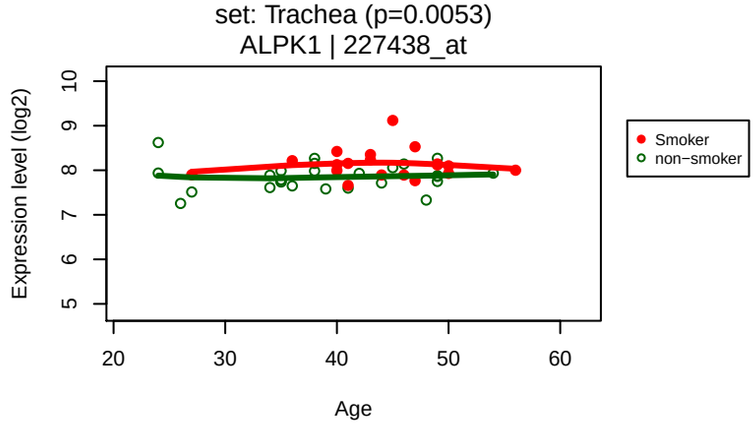

Gene expression ratio in Males
gene: ALPK1 | 227438_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.133
p=0.8559
p=0.108
p=0.7089
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALPK1 | 227438_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3369
p=0.0274
p=0.0069
p=0.088
SmallAirway2
LargeAirway2
Trachea
Macrophage2

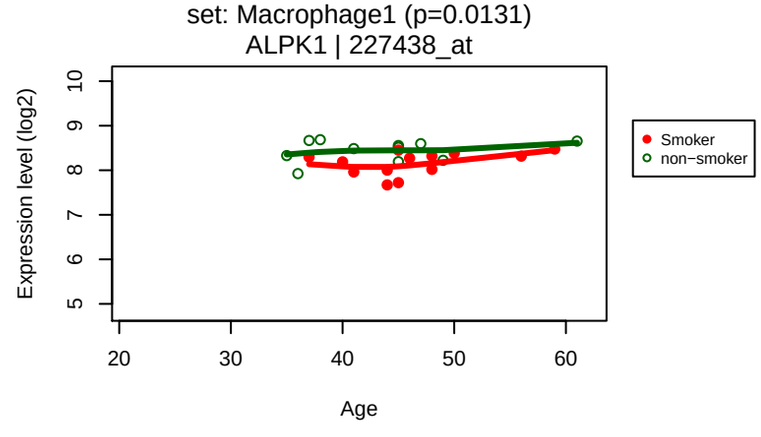

set: SmallAirway1 (p=0.5574)
ALPK1 | 227438_at
Expression level (log2)
Age
Smoker
non−smoker

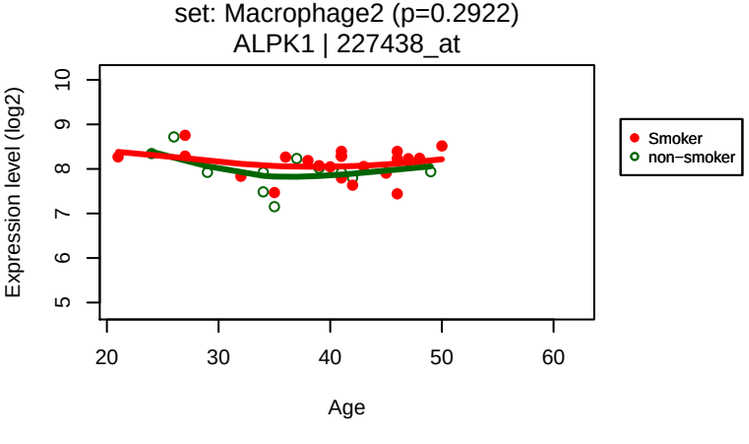

set: SmallAirway2 (p=0.0846)
ALPK1 | 227438_at
Expression level (log2)
Age
Smoker
non−smoker

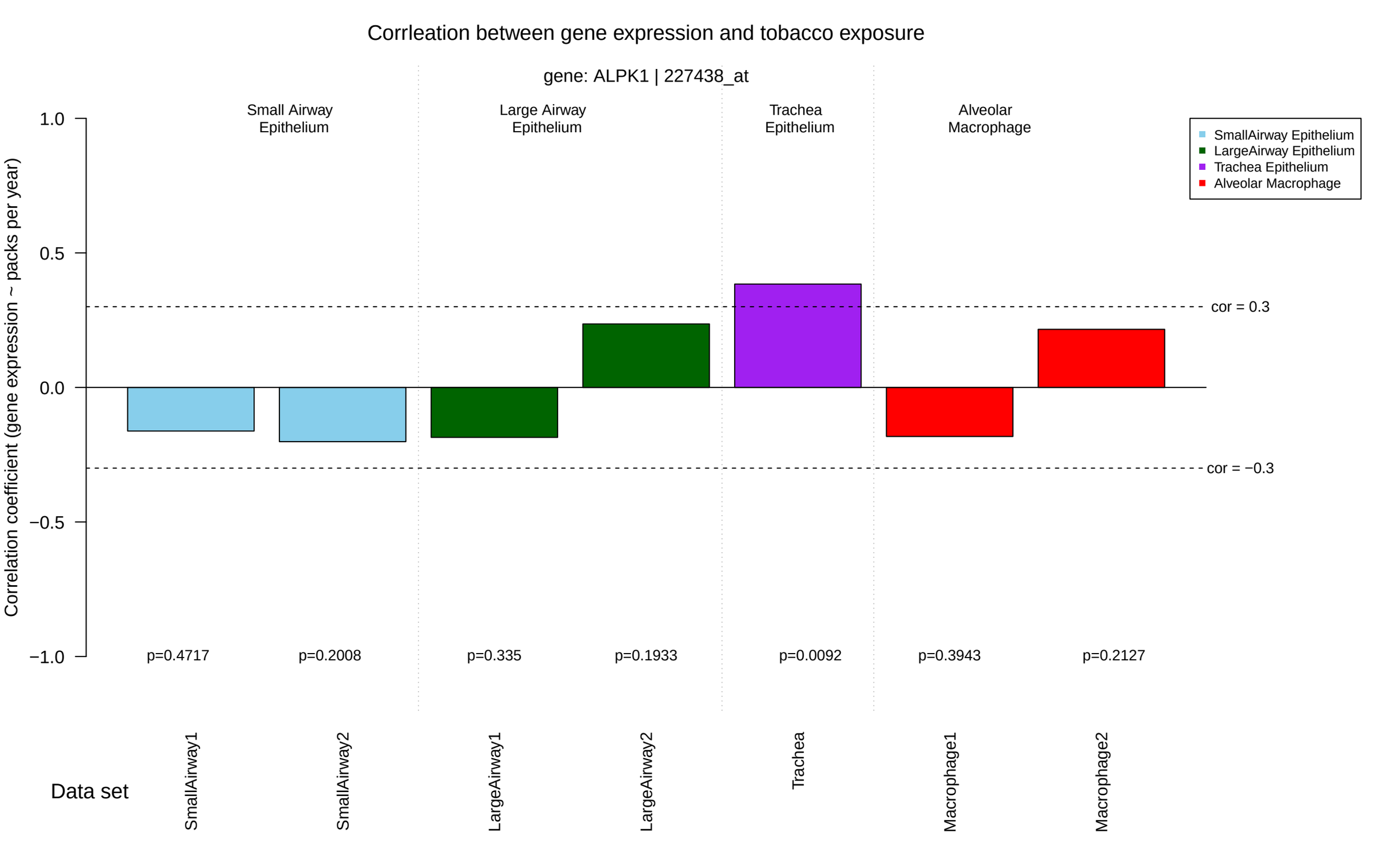

set: LargeAirway1 (p=0.2199)
ALPK1 | 227438_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5239)
ALPK1 | 227438_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0053)
ALPK1 | 227438_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0131)
ALPK1 | 227438_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2922)
ALPK1 | 227438_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALPK1 | 227438_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4717
p=0.2008
p=0.335
p=0.1933
p=0.0092
p=0.3943
p=0.2127
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2